Name: _____

TOPIC TEST 2

Chapter 1: Three-dimensional vectors

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 6 short-answer questions (45 marks).
- Part C: 2 analysis questions (10 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $M(4, -1, -7)$ and which of the following points are on a line parallel with the x-axis?

A $(-4, -1, -7)$

B $(3, -1, 7)$

C $(-4, 1, -7)$

D $(4, -1, 5)$

E $(4, 2, -7)$

[1 mark]

2 The distance between $A(p, q, r)$ and $B(p, s, r)$ is:

A $|r - p|$

B $|q - p|$

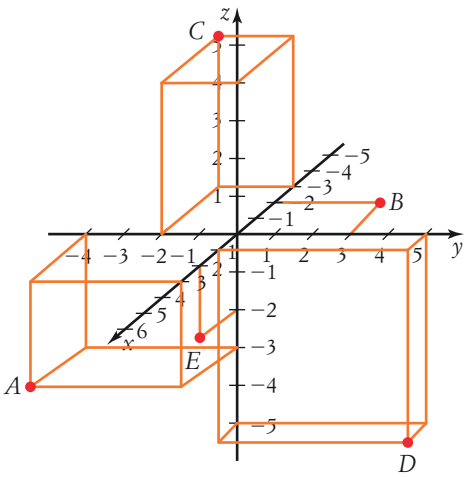
C $|s - q|$

D $|s - r|$

E $|p - q|$

[1 mark]

Use the diagram below to answer Questions 3–5.



3 The point lying on the *xy*-plane is:

A *A*

B *B*

C *C*

D *D*

E *E*

[1 mark]

4 The point *E* is best described as lying on the:

A *yz*-plane

B *y*-plane

C *xz*-plane

D *x*-plane

E *xy*-plane

[1 mark]

5 The distance from the origin to *A* is closest to:

A 3

B 5

C 4

D 6

E 10

[1 mark]

Part B

Short-answer questions

6 short-answer questions
45 marks
Show your working where appropriate.

1 Find the radius and centre of the sphere

$x^2 + y^2 + z^2 - 14x + 4y - 12z + 25 = 0$

[6 marks]

2 Given $\mathbf{a} = (-1, 3, 4)$, $\mathbf{b} = (5, -2, 6)$, and $P = (4, 3, -7)$ and $Q = (-3, -1, 4)$:

a find $-2\mathbf{a} + 5\mathbf{b}$

b show that $(\mathbf{a} - \mathbf{b})$ is parallel to $(\mathbf{b} - \mathbf{a})$

c find $\mathbf{PQ}$ and $|\mathbf{PQ}|$

d find $\hat{\mathbf{b}}$

[10 marks]

3 a Show that the vectors $(-2, -5, 6)$ and $(-1, 4, 3)$ are perpendicular.

b Find the projection of $\mathbf{a} = (-4, 2, -3)$ on $\mathbf{b} = (2, -5, 1)$

c Find the angle between $\mathbf{p} = 3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$ and $\mathbf{q} = 2\mathbf{i} - 6\mathbf{j} - 7\mathbf{k}$.

[10 marks]

4 a Find the vector product of vectors of magnitudes 24 and 17 if the angle between them is 42°.

b Find the vector product of $\mathbf{u} = (7, -3, 2)$ and $\mathbf{v} = (-5, 4, -6)$.

[7 marks]

5 a Find the Cartesian form of the equation of the straight line which passes through the points $(3, 7, -2)$ and $(4, 5, -3)$.

b Calculate the equivalent vector equation for the line.

[6 marks]

6 Find the equation of the plane that contains the points $P(4, -3, 1)$, $Q(5, 3, 2)$ and $R(-6, 2, 5)$.

[6 marks]

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Part C

Analysis questions

2 analysis questions
10 marks
Show your working where appropriate.

1 Given a quadrilateral with opposite sides parallel and equal, prove that the quadrilateral is a parallelogram.

[5 marks]

2 Find an equation for the plane consisting of all points that are equidistant from $P(-1, 2, -3)$ and $Q(4, -2, 2)$.

[5 marks]

[Total marks: 60]

Answers

Part A – Multiple-choice questions [1 mark each]

1 Points on a line parallel to the x-axis will have y- and z-coordinates the same but different x-coordinates.

$\therefore$ **A** **[1 mark]**

2 A and B have different y-coordinates but the x- and z-coordinates are the same.

The points are on a line parallel to the y-axis.

The distance between A and B will be the absolute value of the difference between the y-coordinates i.e. $|s - q|$

$\therefore$ **C** **[1 mark]**

3 Any point on the xy-plane will have a z-coordinate of 0.

$B = (-2, 3, 0)$.

$\therefore$ **B** **[1 mark]**

4 $E = (3, 0, -2)$.

The y-coordinate is 0, E lies on the xz-plane.

$\therefore$ **C** **[1 mark]**

5 Distance from the origin to $A = \sqrt{3^2 + (-4)^2 + (-3)^2} \approx 5.83 \approx 6$

$\therefore$ **D** **[1 mark]**

Part B – Short-answer questions

1 $x^2 + y^2 + z^2 - 14x + 4y - 12z + 25 = 0$

$x^2 + y^2 + z^2 - 14x + 4y - 12z = -25$

$(x^2 - 14x) + (y^2 + 4y) + (z^2 - 12z) = -25$ **[1 mark]**

$(x^2 - 14x + 49) + (y^2 + 4y + 4) + (z^2 - 12z + 36) = 49 + 4 + 36 - 25$ **[1 mark]**

Factorise the squares. $(x - 7)^2 + (y + 2)^2 + (z - 6)^2 = 64$ **[1 mark]**

$(x - 7)^2 + (y + 2)^2 + (z - 6)^2 = 8^2$ **[1 mark]**

Compare with $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$

Centre $= (7, -2, 6)$ **[1 mark]**

Radius $= 8$ **[1 mark]**

[6 marks altogether]

2 **a** $-2\mathbf{a} + 5\mathbf{b} = -2(-1, 3, 4) + 5(5, -2, 6)$ **[1 mark]**

$= (2, -6, -8) + (25, -10, 30)$

$= (27, -16, 22)$ **[1 mark]**

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b $\mathbf{a} - \mathbf{b} = (-1, 3, 4) - (5, -2, 6)$

$= (-6, 5, -2)$ **[1 mark]**

$\mathbf{b} - \mathbf{a} = (5, -2, 6) - (-1, 3, 4)$

$= (6, -5, 2)$

$= -1(-6, 5, -2)$ **[1 mark]**

$= -1(\mathbf{a} - \mathbf{b})$

$(\mathbf{a} - \mathbf{b})$ is parallel to $(\mathbf{b} - \mathbf{a})$ **[1 mark]**

c $\mathbf{PQ} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$

$= (-3 - 4, -1 - 3, 4 - -7)$

$= (-7, -4, 11)$ **[1 mark]**

$|\mathbf{PQ}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

$= \sqrt{(-7)^2 + (-4)^2 + 11^2}$

$= \sqrt{186}$ **[1 mark]**

d $\hat{\mathbf{b}} = \dfrac{1}{|\mathbf{b}|}\mathbf{b}$ **[1 mark]**

$= \dfrac{1}{\sqrt{5^2 + (-2)^2 + 6^2}}(5, -2, 6)$ **[1 mark]**

$= \dfrac{1}{\sqrt{65}}(5, -2, 6)$

$= \dfrac{\sqrt{65}}{65}(5, -2, 6)$ **[1 mark]**

[10 marks altogether]

3 a $(-2, -5, 6) \cdot (-1, 4, 3) = 2 - 20 + 18$

$= 0$ **[1 mark]**

Since the scalar product is zero, the vectors are perpendicular. **[1 mark]**

b $p_{\mathbf{b}}\,\mathbf{a} = \mathbf{a} \cdot \hat{\mathbf{b}}$ **[1 mark]**

$\hat{\mathbf{b}} = \dfrac{1}{|\mathbf{b}|}\mathbf{b}$

$= \dfrac{1}{\sqrt{2^2 + (-5)^2 + 1^2}}(2, -5, 1)$

$= \dfrac{1}{\sqrt{30}}(2, -5, 1)$ **[1 mark]**

$p_{\mathbf{b}}\,\mathbf{a} = (-4, 2, -3) \cdot \dfrac{1}{\sqrt{30}}(2, -5, 1)$

$= \dfrac{1}{\sqrt{30}}(-8 - 10 - 3)$ **[1 mark]**

$= -\dfrac{21}{\sqrt{30}} \approx -3.8$ **[1 mark]**

c $\cos(\theta) = \dfrac{\mathbf{m}.\mathbf{n}}{|\mathbf{m}||\mathbf{n}|}$ **[1 mark]**

$= \dfrac{6 - 12 + 35}{\sqrt{3^2 + 2^2 + (-5)^2} \times \sqrt{2^2 + (-6)^2 + 7^2}}$ **[1 mark]**

$= \dfrac{29}{\sqrt{38} \times \sqrt{89}}$

$= \dfrac{29}{\sqrt{3382}}$ **[1 mark]**

$\theta \approx 60.1°$ **[1 mark]**

[10 marks altogether]

4 a $|\mathbf{v}| = |\mathbf{v}_1|\,|\mathbf{v}_2| \sin(\theta)$ **[1 mark]**

$= 24 \times 17 \times \sin(42°)$ **[1 mark]**

≈ 273 **[1 mark]**

The vector product is about 273 in a direction perpendicular to the plane of the original vectors. **[1 mark]**

b $\mathbf{u} = 7\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}, \mathbf{v} = -5\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$ **[1 mark]**

$\mathbf{u} \times \mathbf{v} = (-3 \times -6 - 2 \times 4)\mathbf{i} - (7 \times -6 - 2 \times -5)\mathbf{j} + (7 \times 4 - -3 \times -5)\mathbf{k}$ **[1 mark]**

$= 10\mathbf{i} + 32\mathbf{j} + 13\mathbf{k}$

$= (10, 32, 13)$ **[1 mark]**

[7 marks altogether]

5 a Let $A(a_1, a_2, a_3) = (3, 7, -2)$

and $B(b_1, b_2, b_3) = (4, 5, -3)$ **[1 mark]**

$$\frac{x - a_1}{b_1 - a_1} = \frac{y - a_2}{b_2 - a_2} = \frac{z - a_3}{b_3 - a_3}$$

$$\frac{x - 3}{4 - 3} = \frac{y - 7}{5 - 7} = \frac{z - (-2)}{-3 - (-2)}$$ **[1 mark]**

$$\frac{x - 3}{1} = \frac{y - 7}{-2} = \frac{z + 2}{-1}$$ **[1 mark]**

b Compare $\dfrac{x - 3}{1} = \dfrac{y - 7}{-2} = \dfrac{z + 2}{-1}$ with $\dfrac{x - a_1}{u_1} = \dfrac{y - a_2}{u_2} = \dfrac{z - a_3}{u_3}$

$a_1 = 3, a_2 = 7, a_3 = -2$

and $u_1 = 1, u_2 = -2, u_3 = -1$ **[1 mark]**

$\mathbf{p} = \mathbf{a} + t\mathbf{u}$

$= a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} + t(u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k})$ **[1 mark]**

$\mathbf{p} = 3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k} + t(\mathbf{i} - 2\mathbf{j} - \mathbf{k})$ **[1 mark]**

[6 marks altogether]

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6 $\mathbf{PR} = (-6 - 4, 2 - (-3), 5 - 1) = (-10, 5, 4)$

$\mathbf{PQ} = (5 - 4, 3 - (-3), 2 - 1) = (1, 6, 1)$ **[1 mark]**

$\mathbf{PR} \times \mathbf{PQ}$ is normal to the plane containing **PR** and **PQ** **[1 mark]**

$\mathbf{PR} \times \mathbf{PQ} = (5 \times 1 - 4 \times 6)\mathbf{i} - (-10 \times 1 - 4 \times 1)\mathbf{j} + (-10 \times 6 - 5 \times 1)\mathbf{k}$

$= -19\mathbf{i} + 14\mathbf{j} - 65\mathbf{k} = \mathbf{n}$ **[1 mark]**

Cartesian equation of the plane is: $a_1(x - x_1) + a_2(y - y_1) + a_3(z - z_1) = 0$

Let $P(4, -3, 1)$, and $\mathbf{n} = (-19, 14, -65)$ **[1 mark]**

$-19(x - 4) + 14(y + 3) - 65(z - 1) = 0$ **[1 mark]**

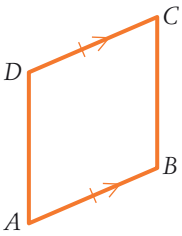
$-19x + 76 + 14y + 42 - 65z + 65 = 0$

$-19x + 14y - 65z + 183 = 0$ is the equation of the plane **[1 mark]**

[6 marks altogether]

Part C – Analysis questions

1 Draw a sketch and label the vertices A, B, C and D.



$|\mathbf{AB}| = |\mathbf{CD}|$ (Given)

$\mathbf{AB} \parallel \mathbf{CD}$ (Given) **[1 mark]**

Thus $\mathbf{AB} = \mathbf{CD}$ **[1 mark]**

$\mathbf{b} - \mathbf{a} = \mathbf{d} - \mathbf{c}$ **[1 mark]**

Rearrange: $\mathbf{d} - \mathbf{a} = \mathbf{c} - \mathbf{b}$

$\mathbf{AD} = \mathbf{BC}$ **[1 mark]**

$ABCD$ is a parallelogram. **[1 mark]**

[5 marks altogether]

2 The midpoint of P and Q lies on the plane. **[1 mark]**

Midpoint of P and Q $= \frac{1}{2}[(-1, 2, -3) + (4, -2, 2)]$

$= \frac{1}{2}(3, 0, -1)$ **[1 mark]**

A normal vector to the plane is given by **QP**

i.e. $4\mathbf{i} - 2\mathbf{j} + 2\mathbf{k} - (-\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$

$= 5\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$ **[1 mark]**

Hence, an equation of the plane is given by:

$5\left(x - \frac{3}{2}\right) - 4(y - 0) + 5\left(z + \frac{1}{2}\right) = 0$ **[1 mark]**

$5x - 4y + 5z - 5 = 0$ **[1 mark]**

[5 marks altogether]

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